

Compulsory Task 2

Microsoft's Cortana

Microsoft's Cortana is a virtual assistant that utilises advanced natural processing (NLP) to understand and respond to both voice and text commands from a user. Cortana is also integrated with different Microsoft products, such as Outlook and Teams. This could potentially help and improve users' efficiency during work or other activities (Microsoft, 2023)

What does it do?

Cortana helps users manage their tasks, setting reminders, sending emails, or even creating calendar entries or events. According to TechTarget, 2021, it aims to streamline daily activities by suggesting insights based on user activity and managing schedules.

Cortana has the ability to manage email communication. In Microsoft's Outlook, it helps users prioritise emails by providing smart suggestions based on said emails context, sentiments, and urgency, similar to what Gmail's auto-response feature does. It can suggest responses such as 'Yes, I am available' or 'That does not fit my current schedule'. It can also help with suggesting follow-up actions or reminders for pending tasks (BotPenguin, 2023).

How does it work?

Cortana relies on several NLP techniques to interpret and respond to the users inputs and requests. It processes spoken or text-based inputs. It uses automatic speech recognition to convert speech to text and then uses NLP to determine the intent and reason behind the request. Once the reason and intent are understood, Cortana formulates an appropriate response.

Let's look at email management: Cortana can utilise contextual analysis by analysing the content of the metadata of incoming emails (like sender, subject, dates, etc.) to suggest relevant actions. It can also learn from previous interactions to improve future suggestions, adapting to the user's preferences and patterns of communications (ITPro Today, 2018).

Cortana is also integrated into Microsoft's cloud-based platform. This allows it to gather insights from multiple sources, like emails or calendar entries, and provide personalised suggestions. This makes Microsoft's assistant to be not only reactive but also anticipatory, providing the user with what they might need before even asking (BotPenguin, 2023).

References

- Microsoft. (2023). *End of support for Cortana*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/topic/end-of-support-for-cortana-d025b39f-ee5b-4836-a954-0ab646ee1efa> [Accessed 4 Mar. 2025].
- TechTarget. (2021). *What is Microsoft Cortana? Everything You Need to Know*. TechTarget. Available at: <https://www.techtarget.com/searchenterprisedesktop/definition/Cortana> [Accessed 4 Mar. 2025].
- BotPenguin. (2023). *Cortana: Features, Uses, Benefits & Limitations*. BotPenguin. Available at: <https://botpenguin.com/glossary/cortana-virtual-assistant> [Accessed 4 Mar. 2025].
- ITPro Today. (2018). *Cortana Cred Rises through Use of Natural Language Processing*. ITPro Today. Available at: <https://www.itprotoday.com/ai-machine-learning/microsoft-aims-to-raise-cortana-s-business-cred-through-the-use-natural-language> [Accessed 4 Mar. 2025].